

ARCH 594/508: AI Research Seminar/Studio—AI in Architecture
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Software Introduction and Set-Up: Python

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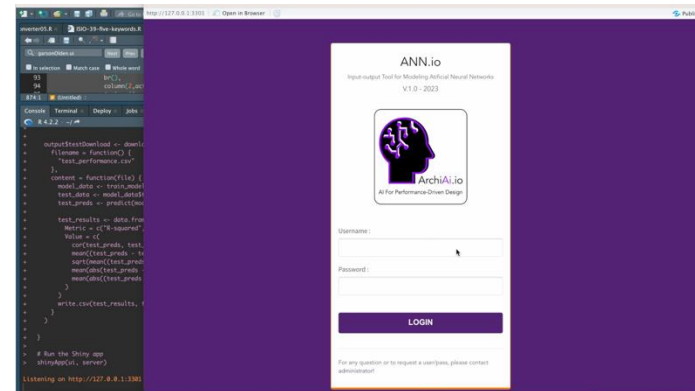
TODAY'S AGENDA

- Introduction to Python
- Key Features of Python
- Popular Python Libraries
- Installing Python Locally
- Introduction to Python using Google Colab
- Hands-on Practice with Google Colab
- Other Software Programs

TECHNOLOGY REQUIREMENTS

In this course we will use these software programs and tools:

- Python programming language (main platform)
- ArchiAi.io



WHAT IS PYTHON?

- Python is a high-level, general-purpose programming language created by Guido van Rossum.
- It was first released in 1991 and is widely used in web development, AI, and more.
- Python emphasizes code readability and simplicity.



```
1  class CodeExample():
2      def printStatement(self):
3          print('Hello World!')
4
5  def main():
6      classEx = CodeExample()
7      classEx.printStatement()
8  if __name__ == "__main__":
9      main()
```

WHY LEARN PYTHON?

- Ease of learning due to simple syntax.
- Versatility in applications such as AI, ML, web development, automation.
- Strong community and library support like NumPy, Pandas, TensorFlow.

Machine Learning Example with Scikit-learn

```
from sklearn.linear_model import LinearRegression

# Example data
X = [[1], [2], [3], [4]]
y = [1, 2, 3, 4]

# Create a LinearRegression model
model = LinearRegression()
model.fit(X, y)

# Make a prediction
prediction = model.predict([[5]])
print(f"Prediction for 5: {prediction[0]}")
```

COMMON PYTHON APPLICATIONS

- Web development: Using frameworks like Django and Flask.
- Data science: Libraries like Pandas, NumPy, and TensorFlow.
- Automation and scripting.
- Scientific computing and academic research.

CORE PYTHON LIBRARIES FOR DATA SCIENCE & AI

- **NumPy**: High-performance array computing and mathematical operations.
- **Pandas**: Tabular data manipulation, preprocessing, and analysis.
- **Matplotlib / Seaborn**: Data visualization—statistical and exploratory plots.
- **Scikit-learn**: Classical machine learning—modeling, evaluation, and pipelines.
- **TensorFlow / PyTorch**: Deep learning frameworks for neural networks and large-scale AI systems.



PYTHON VS. OTHER LANGUAGES

- **Python vs Java:** Simpler syntax but slower performance.
- **Python vs C++:** Easier to learn, but C++ offers better performance.
- **Python vs R:** Python is a general-purpose language, while R is mainly used for statistics and data analysis.

INSTALLING PYTHON LOCALLY



- Go to the official Python website: <https://python.org>
- Download the correct version for your operating system (Windows, macOS, Linux).
- Run the installer and follow the setup instructions.
- Make sure to check "Add Python to PATH" during installation.
- After installation, verify by typing `python --version` in the terminal or command prompt.

INSTALLING PYTHON LIBRARIES WITH ANACONDA

- Download and install Anaconda from <https://www.anaconda.com/products/distribution>
- Anaconda is a distribution that comes with Python and many popular libraries pre-installed.
- You can use the Conda package manager to install additional libraries.
- Example to install a library: Open terminal/command prompt and type `conda install numpy`.
- You can also create isolated environments using `conda create --name myenv`.



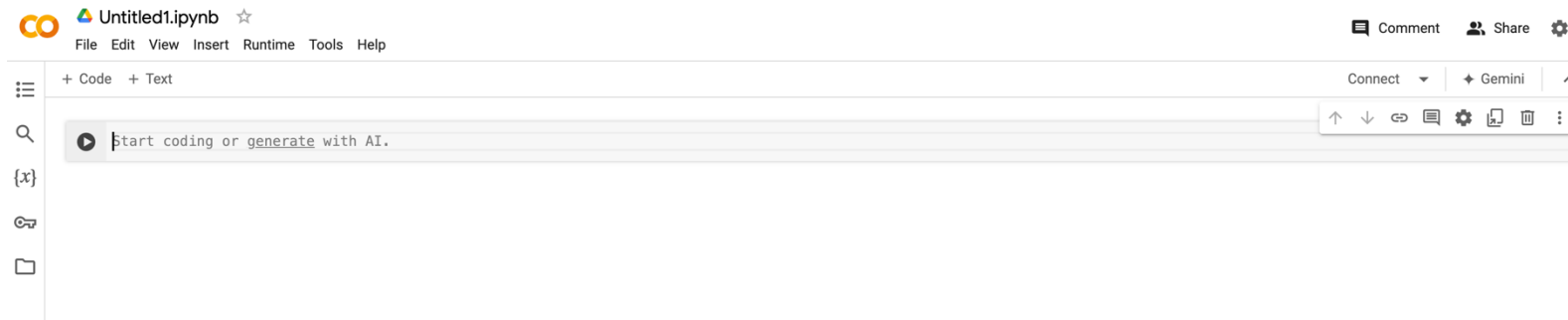
WHAT IS GOOGLE COLAB?

- Google Colab is a free cloud service for Python development.
- No local setup is needed, runs on Jupyter Notebooks.
- Comes with pre-installed libraries such as NumPy, Pandas, TensorFlow.
- Ideal for data science and machine learning.
- Easy to share and collaborate.



GETTING STARTED WITH GOOGLE COLAB

- Go to <https://colab.research.google.com>
- Sign in with your Google account.
- Create a new notebook to start writing and running Python code.
- Colab provides free access to computing resources, including GPUs.



CONNECTION WITH GOOGLE DRIVE

- Colab integrates with Google Drive, allowing you to save and access your projects easily from the cloud. This ensures your work is automatically saved and can be accessed from any device.



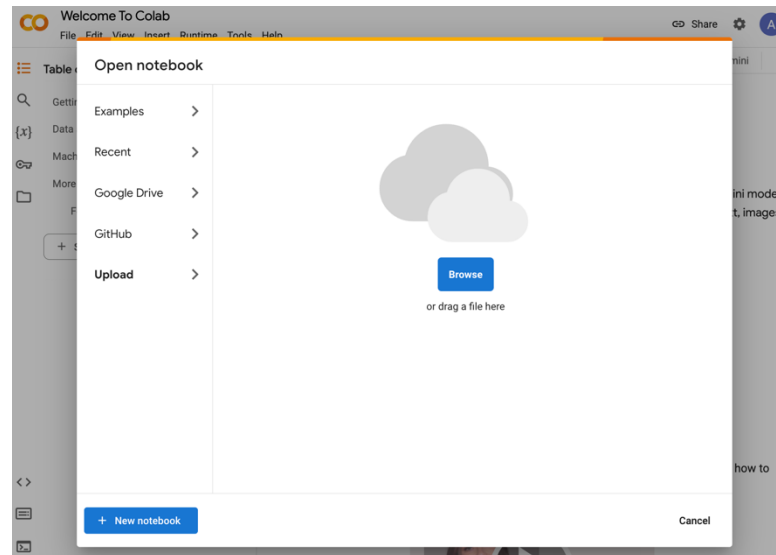
GETTING STARTED WITH GOOGLE COLAB

- In this session we will introduce the python coding language in a Jupyter notebook using [Google Colab](#).



GOOGLE COLAB TUTORIAL SET-UP

1. Open up a web browser. Google “google colab” click on the first entry. You may also access the website by following <https://colab.research.google.com/>
2. Go to Canvas and download the file called “00_AI_Studio_Python Basics.ipynb”
3. You will be prompted with an *Open notebook* popup. Click “Upload” on the bottom of the menu list. Browse for and upload the notebook that you downloaded in Step 2.



CONCLUSION

- Python is versatile, powerful, and easy to learn.
- Google Colab provides a free environment for Python development.
- Get started by exploring Python libraries and hands-on coding in Colab.

PYTHON RESOURCES

- Please see the following resources for more introductory information:
 - Whirlwind Tour Of Python - great book with free tutorials by a UW alum
 - <https://jakevdp.github.io/WhirlwindTourOfPython/>
 - Basic webpage tutorial
 - <https://www.dataquest.io/blog/jupyter-notebook-tutorial/>
 - Python basics tutorial made by google research, more comprehensive
 - https://colab.research.google.com/github/data-psl/lectures2020/blob/master/notebooks/01_python_basics.ipynb
- Other tips
 - Things can get complicated kind of quickly. Google will be your best friend if you need syntax help!
 - When searching for resources on google - include the library that you are working with in your search term.
 - For example: "matrices in numpy", "rename dataframe columns in pandas", "merge dataframes in pandas".

NEXT CLASS

- **Lecture:** Data in Architecture & Exploratory Data Analysis

